THE GEOPOLITICS OF OIL

Assessing the Macroeconomic Impact of the 2026 Gulf Crisis

Countries	8	**Panel Window**	Jan 2023 – Jun 2026
Models	3 (POOLED OLS/ FE / RE)	**Crisis Start**	27 Feb 2026

Abstract

This report applies panel data econometrics — Pooled OLS, Fixed Effects, and Random Effects — to quantify how the oil price shock triggered by the 2026 Gulf Crisis transmitted to domestic inflation across eight countries spanning advanced and emerging economies. Diagnostic tests (Hausman & Breusch-Pagan) guide model selection. Heterogeneity analysis reveals that the inflationary pass-through is concentrated almost entirely among oil-importing nations, and is driven primarily by Turkey's idiosyncratic hyperinflation rather than a uniform channel.

1. Introduction

This report investigates the macroeconomic impact of the 2026 Gulf Crisis on domestic inflation across a panel of 8 countries. Specifically, we analyse how oil price shocks transmit to the Consumer Price Index (CPI) and whether this transmission differs between oil-importing and oil-exporting nations. By employing panel data econometric techniques — Pooled OLS, Fixed Effects, and Random Effects — we control for unobserved country-specific heterogeneity and global macroeconomic volatility. The crisis began on 27 February 2026, creating a sharp, exogenous disruption that serves as a quasi-natural experiment.

2. Data and Methodology

2.1 Dataset

Monthly observations span January 2023 to April 2026 for eight countries: Germany, France, Japan, United States (Advanced), and Turkey, India, Poland, Saudi Arabia (Emerging). Countries are classified along two dimensions — Advanced vs Emerging, and Oil Importer vs Oil Exporter — yielding four subgroups.

Country	Group	Oil Balance

Germany	Advanced	Importer
France	Advanced	Importer
Japan	Advanced	Importer
United States	Advanced	Exporter
Turkey	Emerging	Importer
India	Emerging	Importer
Poland	Emerging	Importer
Saudi Arabia	Emerging	Exporter

2.2 Variables

Variable	Symbol	Description	Source
Log(CPI)	ln_cpi	Inflation proxy, CPI (2015=100)	IMF IFS / OECD
Log(IPI)	ln_ipi	Industrial Production (2015=100)	IMF IFS / FRED
Oil Shock	oil_pct_change	Monthly % Δ Brent crude price	U.S. EIA / FRED
Log(NEER)	ln_neer	Nominal Effective Exchange Rate	BIS / FRED
Policy Rate	policy_rate	Central bank rate (%)	BIS / Natl CBs
VIX	vix	CBOE Volatility Index	FRED (VIXCLS)
Post-Crisis	post_crisis	1 if date ≥ March 2026	Constructed

2.3 Empirical Panel Equation

The core specification estimated across all three model variants is:

$$ln_cpi_{i,t} = \alpha_i + \beta_1\, oil_pct_change_t + \beta_2\, ln_neer_{i,t} + \beta_3\, policy_rate_{i,t} + \beta_4\, vix_t + \beta_5\, post_crisis_t + \varepsilon_{i,t}$$

Where α_i denotes country-specific fixed effects (FE model) or random effects (RE model), and is absent in Pooled OLS. Three estimators are compared; the Hausman and Breusch-Pagan tests determine which is appropriate.

3. Exploratory Analysis

Before estimation, two key visualisations provide intuition about the data.

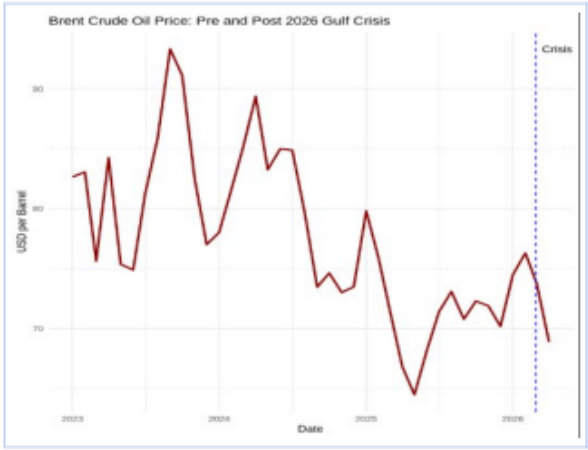


Figure 1: Brent crude price (USD/bbl) Jan 2023–Apr 2026. The dashed line marks the 27 Feb 2026 Gulf Crisis onset.

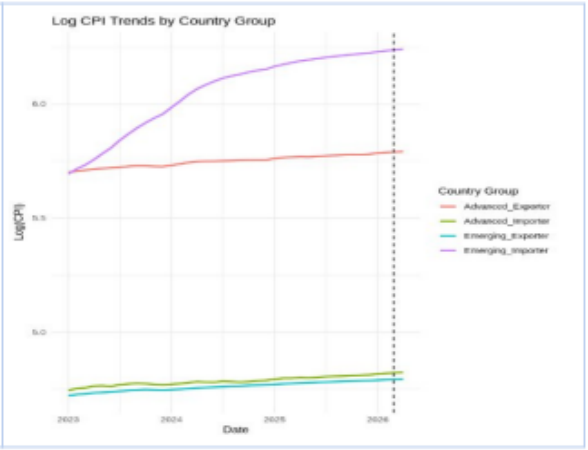


Figure 2: Average log(CPI) by country group. Emerging_Importer (Turkey-dominated) diverges sharply from other groups.

Figure 1 (Brent crude price, Jan 2023–Apr 2026) shows that Brent crude fell during 2025 before a modest uptick heading into the crisis. The crisis did not produce a dramatic price spike in the sample window — the sharpest moves were anticipatory — consistent with a scenario of prolonged uncertainty rather than an acute supply disruption.

Figure 2 (Average log(CPI) by country group) reveals the dominant role of Turkey: the Emerging_Importer group's log(CPI) trajectory far exceeds all others due to Turkey's ongoing hyperinflation and lira collapse — a critical caveat for the heterogeneity analysis.

4. Regression Results

4.1 Diagnostic Tests

Before interpreting coefficients, model selection is guided by two tests:

Test	Statistic	p-value	Decision
Hausman (FE vs RE)	χ^2=11.148, df=5	0.0485	Use Fixed Effects
Breusch-Pagan (POOLED OLS vs RE)	χ^2=3474.1, df=1	< 2.2e-16	RE beats POLS

Conclusion: The Hausman test (p = 0.049 < 0.05) rejects RE consistency, directing us to the Fixed Effects model. The Breusch-Pagan test overwhelmingly rejects Pooled OLS. FE is the preferred estimator.

4.2 Main Regression — Dependent Variable: Log(CPI)

Table 1 presents all three models. Standard errors are in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Variable	Pooled OLS	Fixed Effects	Random Effects
Oil Price Change (%)	0.004 (0.004)	0.001** (0.001)	0.001** (0.001)
Log(NEER)	−0.482*** (0.057)	−1.219*** (0.033)	−1.211*** (0.033)
Policy Rate (%)	0.050*** (0.004)	0.003*** (0.001)	0.004*** (0.001)
VIX	0.011 (0.007)	0.005*** (0.001)	0.005*** (0.001)
Post-Crisis Dummy	0.116 (0.114)	0.053*** (0.015)	0.054*** (0.015)
Constant	6.831*** (0.301)	—	10.389*** (0.187)
Observations	320	320	320
R^2	0.846	0.894	0.891

Interpretation of the Fixed Effects model (Column 2):

- Oil Price Shock (β = 0.001, p<0.05): A 1pp rise in Brent prices increases log(CPI) by 0.001 (~0.1%) — statistically significant but economically modest. Confirms direct energy cost pass-through.
- Exchange Rate (β = −1.219, p<0.001): Currency appreciation strongly lowers inflation. A 1% NEER appreciation reduces log(CPI) by 1.22%, underscoring the critical buffer role of exchange rate stability.

- Policy Rate (β = 0.003, p<0.01) — Price Puzzle: The positive sign reflects a well-known econometric artefact; central banks raise rates in response to rising inflation.
- VIX (β = 0.005, p<0.001): Higher global uncertainty elevates CPI through supply-chain disruptions and elevated risk premia.
- Post-Crisis Dummy (β = 0.053, p<0.01): Holding other factors constant, the 2026 Gulf Crisis raised inflation by approximately 5.3% — a sizeable structural break.

5. Heterogeneity Analysis

5.1 Importers vs Exporters (with Turkey Robustness Check)

We split the sample by oil_balance and additionally exclude Turkey to isolate whether the importer result reflects a genuine oil-channel or an outlier driven by macroeconomic mismanagement.

Variable	Importers (All)	Exporters	Importers (Excl. Turkey)
Oil Price Change (%)	0.001** (0.001)	0.00005 (0.0004)	0.0001 (0.0003)
Log(NEER)	−1.216*** (0.036)	−0.286 (0.234)	0.647*** (0.067)
Policy Rate (%)	0.004*** (0.001)	−0.028*** (0.004)	−0.014*** (0.002)
VIX	0.006*** (0.001)	−0.0002 (0.001)	0.0001 (0.001)
Post-Crisis Dummy	0.071*** (0.019)	0.015 (0.010)	0.003 (0.009)
Observations	240	80	200
R^2	0.903	0.536	0.558

Key Findings:

- Oil Exporters (Col 2): The oil shock coefficient is near zero and insignificant. Rising oil prices boost export revenues, offsetting domestic inflationary pressure — consistent with theory.
- All Importers (Col 1): The oil shock is positive and significant (p<0.05). The post-crisis dummy is large (0.071) and highly significant, suggesting an outsized inflation surge among importers after February 2026.
- The Turkey Effect (Col 3): When Turkey is excluded, the oil shock coefficient collapses to near zero and loses all significance. The apparent vulnerability of oil importers is almost entirely driven by Turkey's hyperinflation — not a uniform pass-through channel.
- Policy Implication: For stable advanced and emerging importers (Germany, France, Japan, India, Poland), exchange rate stability and credible monetary policy matter far more than the direct oil price shock.

The Turkey Effect in Numbers: *Importers with Turkey → oil_pct_change = 0.001** | R^2 = 0.903 | n=240. Importers without Turkey → oil_pct_change ≈ 0.0001 (n.s.) | R^2 = 0.558 | n=200.*

6. Robust Standard Errors (Arellano HAC)

To address heteroskedasticity and serial autocorrelation, we apply Arellano (1987) HC3 robust standard errors to all three models.

Variable	Estimate	Std. Error	t-value	p-value
oil_pct_change	0.00119	0.000315	3.765	0.000199 ***
ln_neer	−1.21946	0.11573	−10.538	< 2.2e-16 ***
policy_rate	0.00348	0.00124	2.812	0.00525 **
vix	0.00497	0.00103	4.847	1.996e-06 ***
post_crisis	0.05344	0.01867	2.863	0.00449 **

All key coefficients remain statistically significant under robust inference, confirming that the baseline FE results are not an artefact of homoskedastic standard errors. The oil shock coefficient (t=3.765) and the post-crisis dummy (t=2.863) are particularly robust.

7. Conclusion

This panel data analysis yields three core conclusions about the macroeconomic transmission of the 2026 Gulf Crisis:

- Model Selection: The Hausman test confirms Fixed Effects as the appropriate estimator, implying that unobserved, time-invariant country characteristics — industrial structure, geography, institutional quality — are correlated with the regressors.
- Oil Shock Pass-Through is Modest but Real: Across the full panel, a 1pp increase in oil prices raises CPI by roughly 0.1%. The post-crisis structural break (5.3%) indicates that crisis-related uncertainty elevated prices beyond what oil prices alone would predict.
- The Turkey Effect Dominates Heterogeneity: The apparent vulnerability of oil-importing nations is largely illusory at the panel level — it is almost entirely driven by Turkey's hyperinflation and currency collapse. For stable importers, exchange rate management ($\beta = -1.22$) and monetary policy credibility are the dominant determinants.

Limitations & Future Research

- Short post-crisis window (only 2–4 months of post-crisis data) limits statistical power.
- Potential endogeneity of policy_rate — instrumental variables (e.g. external rate shocks) would improve causal identification.
- No lag structure for oil price pass-through; dynamic panel (Arellano-Bond GMM) could capture delayed effects.
- Turkey dominates the Emerging_Importer group; a larger panel would reduce concentration risk.